Predicting NBA Career Outcomes from Early-Career Performance

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2026-04-15

1 Abstract

Can a player's first few NBA seasons reliably predict where their career will end up? We use public NBA data accessed through `hoopR`, combining early-career box score statistics with draft information, to model career outcomes across four tiers: `Out of League`, `Role Player`, `Starter`, and `All-Star`. This structure captures more nuance than a simple "bust or not" framing, and allows us to ask how much signal exists in the first one to three seasons. To answer this question, we fit a multinomial logistic regression, a Bayesian multinomial model, and a random forest to predict career outcomes. Results suggest that early scoring, playing time, and draft position are the strongest predictors of a player's career trajectory.

2 Introduction

One of the central challenges in basketball personnel decisions is evaluating players before their careers have fully developed. NBA teams must make roster and contract commitments based on limited data and the cost of being wrong (giving up on a future star too early or investing heavily in a player who never breaks out) is enormous. For example, the Clippers traded away SGA for Paul George, which proved to be a big blunder. This pressure is sharpest at the top of the draft. Missing on a high pick is one of the most expensive mistakes a front office can make, because those picks carry both the highest expectations and the highest opportunity cost. This raises a fundamental prediction question: how much does a player's first one to three seasons in the league actually tell us about where their career will end up?

The question is harder than it looks. Early NBA performance is a noisy signal. Rookies vary enormously in how ready they are to contribute right away, and the circumstances surrounding a young player such as their team, their role, and competition for minutes shape their statistics in ways that may not reflect their true talent. Ben Simmons is a clear example: through his first three seasons he was an elite facilitator and defender, averaging over 15 points, 8 rebounds, and 8 assists per game and earning an All-Star selection. By conventional early-career metrics he looked like would be a key player for the Sixers. Yet his career derailed dramatically due to several factors including injuries, work ethic, and refusing to shoot. His early statistics were genuinely impressive but also genuinely misleading about his long-run ceiling. Cases like his are a reminder that early-career prediction is uncertain and that even high draft position combined with strong early numbers is no guarantee of a sustained career.

Despite that uncertainty, the question is worth asking quantitatively. Our project models career outcome tiers: `Out of League`, `Role Player`, `Starter`, or `All-Star` as a function of early-career performance averages and draft position. Draft pick is included as a predictor both because it captures pre-NBA expectations and because it is correlated with the opportunities and roles a young player receives. We use three modeling approaches that were motivated by class examples: multinomial logistic regression for interpretable coefficient-based inference, a Bayesian multinomial model for full posterior uncertainty quantification, and a random forest as a flexible nonlinear benchmark. Together they let us assess both what predicts career outcomes and how confident we should be in those predictions given the noisiness of early-career data.

3 Data

We use public NBA player data accessed through `hoopR`, pulling season-level box score statistics for all players who appeared in at least ten games in any season from 2000 through 2015. The ten-game floor removes injury-shortened appearances that would otherwise introduce noise into early-career averages. For each player, we identify their debut season and retain only their first one to three NBA seasons, then compute per-player averages across those seasons: points, rebounds, assists, steals, blocks, field goal percentage, three-point percentage, free throw percentage, and minutes per game. We also record how many seasons were observed (`n_seasons`), since some players have only one or two early seasons available in the window.

Career outcome labels are constructed from each player's full career totals, also pulled via `hoopR`. We define the four tiers as follows: **All-Star** requires a career scoring average above 18 points per game sustained over at least eight approximate seasons; **Starter** requires either a career minutes average above 28 per game or sustained scoring above 14 points per game over at least six seasons; **Role Player** requires at least 15 career minutes per game over four or more seasons; and everyone else is labeled **Out of League**. These thresholds are approximations of career success, which is itself unclear. In hindset, there were limitations with this approach that we will discuss later. Draft position is joined from player biographical data, also via `hoopR`, and players without a recorded draft number (undrafted players) are excluded, leaving one row per player in the final modeling table.

The first plot below shows how draft position relates to long-run outcome. Earlier picks are clearly associated with better outcomes, but the overlap is substantial: many lottery picks end up as Role Players or Out of League, and some late picks become Starters. This overlap shows how draft position is informative but far from deterministic, and the question is whether early-career performance adds predictive signal on top of it.

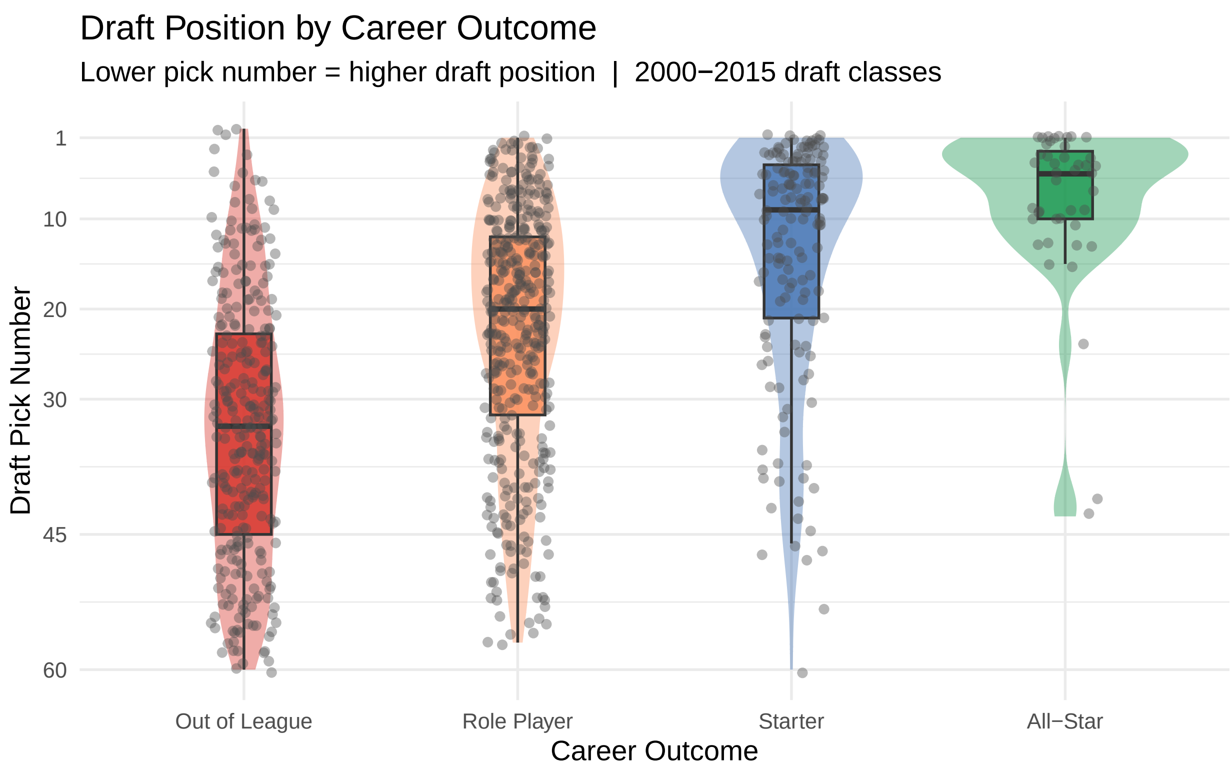

Figure 1: Draft position by career outcome. Lower pick number = higher draft position. The substantial overlap across tiers motivates using early-career statistics alongside draft pick as predictors.

Early scoring production separates outcome tiers more cleanly. The density plot below shows that All-Stars and Starters score substantially more in their first three seasons than Role Players or players who wash out, with relatively little overlap at the extremes. At the same time, the distributions are not perfectly separated where moderate early scorers appear across all four outcome tiers, which is consistent with the idea that early production is a useful but imperfect signal.

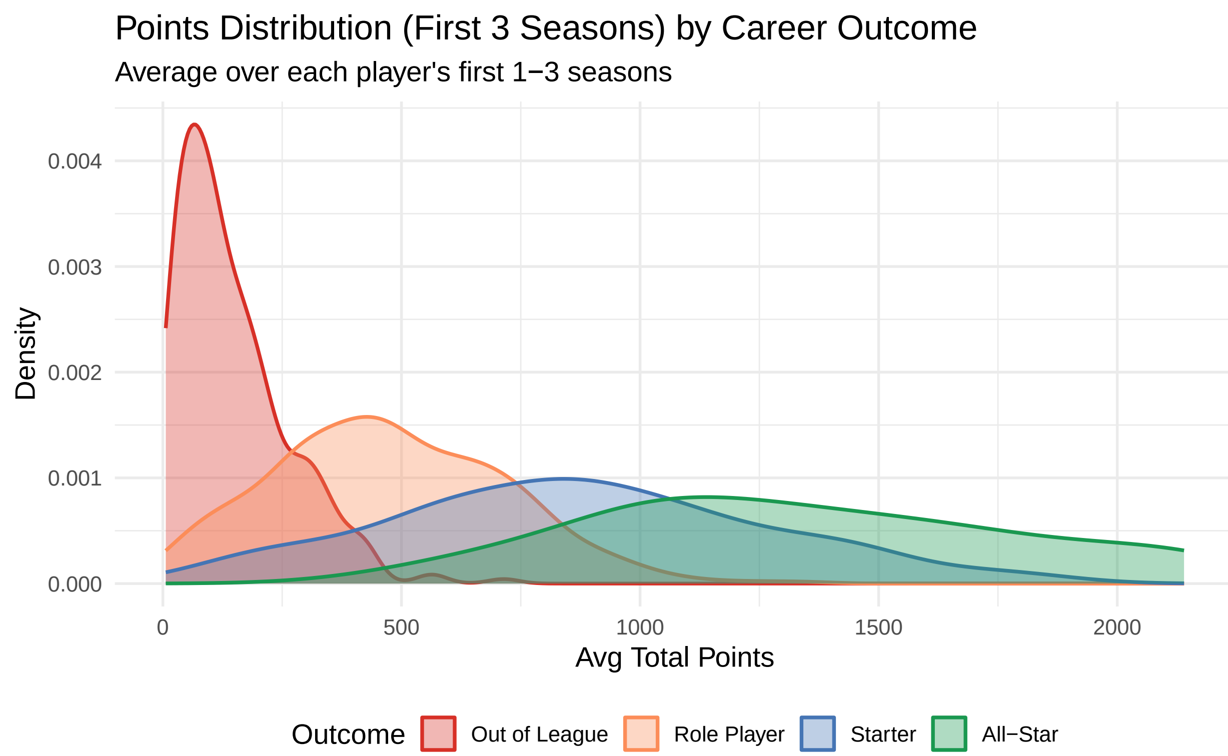

Figure 2: Distribution of average points per game in the first 1–3 seasons by career outcome tier. Better outcomes are associated with higher early scoring, though the distributions overlap considerably in the middle.

Finally, the trajectory plot below shows how scoring evolves across the first three seasons for a sample of players within each tier. On average, All-Stars and Starters show increasing or steadily high scoring, while players who eventually wash out tend to score less and show flatter or declining trajectories. However, the individual variation within each tier is large, reinforcing that early-career statistics are probabilistic guides rather than firm predictors of career outcomes.

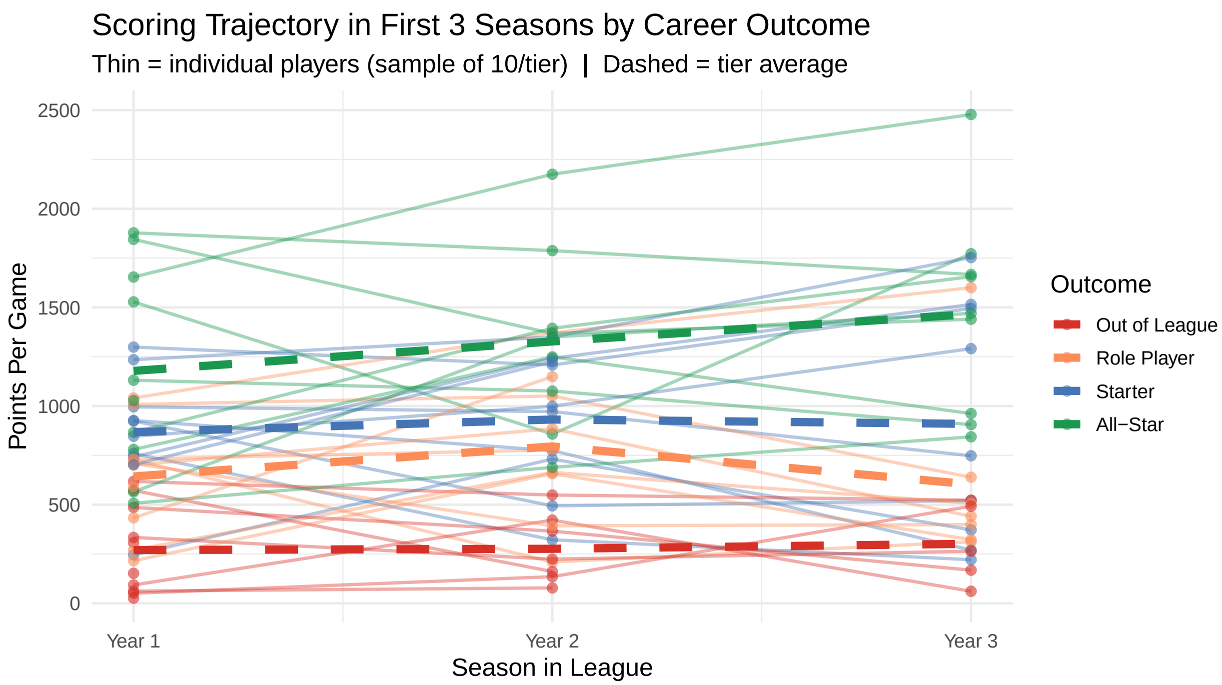

Figure 3: Scoring trajectories in the first 3 seasons for a sample of players per outcome tier. Dashed lines show the tier average. Within-tier variation is substantial across all groups.

4 Methods

4.1 Jiwon Park - Multinomial Logistic Regression

The first model is a multinomial logistic regression model for the four-level career outcome. Let Y_i denote the career tier for player i. The model assumes

$$P(Y_i = k \mid x_i) = \frac{\exp(\alpha_k + x_i^\top \beta_k)}{\sum_{j=1}^{K} \exp(\alpha_j + x_i^\top \beta_j)},$$

with `Out of League` used as the baseline class. This approach is well-suited to the problem because the response is inherently multiclass and the model gives direct coefficient-based interpretation: each coefficient describes how a one-unit change in an early-career statistic shifts the log-odds of landing in a given outcome tier relative to washing out of the league entirely.

The predictors are the full set of early-career averages and draft information: `draft_pick`, `avg_ppg`, `avg_rpg`, `avg_apg`, `avg_spg`, `avg_bpg`, `avg_fg_pct`, `avg_fg3_pct`, `avg_ft_pct`, `avg_min_pg`, and `n_seasons`. The model is fit on an 80% training split and evaluated on the held-out 20% using accuracy and a confusion matrix across the four outcome classes.

Uncertainty is quantified with 500-resample bootstrap confidence intervals on all coefficients. This is appropriate because the main quantities of interest are coefficient directions and magnitudes, and because the bootstrap naturally captures sampling variability in the training data without making distributional assumptions about the coefficient estimates.

4.2 Andy Ouyang - Bayesian Logistic Model

Next, we fit a Bayesian multinomial logistic regression using Stan. The model uses the same multiclass outcome structure as the multinomial logistic regression, but estimates full posterior distributions over all parameters rather than point estimates. This is valuable for the career prediction problem because many players sit near the boundaries between outcome tiers, and a posterior distribution makes that uncertainty explicit rather than collapsing it into a single predicted class. Similar to the previously specified frequentist model, `Out of League` is used as the baseline class.

To keep computation stable and the model interpretable, a reduced predictor set is used: `draft_pick`, `avg_ppg`, `avg_rpg`, `avg_apg`, `avg_min_pg`, and `avg_fg_pct`. All coefficients and intercepts receive weakly informative Normal$(0, 2.5)$ priors, which regularize extreme estimates without strongly constraining the posterior. This also helps with stabilizing estimation given the correlation and noise that is present in early-career performance data. Additionally, all predictors were standardized prior to modeling to improve overall stability and make the priors comparable across coefficients. Four chains are run with 2000 iterations each (1000 warmup), and convergence is verified using R-hat diagnostics. Evaluation combines held-out predictive accuracy and macro-F1 to account for class imbalance across outcome tiers, along with posterior class probability distributions for illustrative player profiles.

As such, we model the outcome using a Bayesian multinomial logistic regression. For player i:
$P(Y_i = k \mid x_i) = \frac{\exp(\alpha_k + x_i^T \beta_k)}{\sum_{j=1}^{K} \exp(\alpha_j + x_i^T \beta_j)}$
with Out of League as the baseline class and weakly informative priors on all parameters:
$\alpha_k \sim \mathcal{N}(0, 2.5), \quad \beta_k \sim \mathcal{N}(0, 2.5)$

This Bayesian multinomial logistic regression assumes (i) conditional independence of observations given predictors, (ii) a linear relationship between the predictors and the log-odds of each outcome class, (iii) a correctly specified multinomial likelihood for the categorical career outcome, and (iv) that predictors are on a comparable scale due to standardization, allowing the weakly informative Normal(0, 2.5) priors to be applied consistently across coefficients. We also assume no perfect multicollinearity among predictors, and that the selected performance metrics capture sufficient signal for distinguishing between outcome tiers.

Uncertainty is quantified directly through posterior credible intervals and posterior predictive probabilities. This extends beyond what a frequentist model provides: rather than a point prediction with a separate confidence interval, the model outputs a full probability distribution over career outcomes for any given player profile.

4.3 Stanley Ou - Random Forest

Lastly, we fit a random forest classifier using the `ranger` package. The motivation for using a tree-based model here is that it could capture how scouts and front offices actually think about young players, in terms of their numbers. A tree can naturally express logic like "if a rookie averages above 25 minutes per game and scores at least 14 points, they look like a Starter trajectory; otherwise check whether efficiency is high enough to support a bench role." Trees also handle the four-tier outcome structure directly and naturally.

Since decision trees tend to overfit, we used a random forest with an ensemble of 500 trees. Each tree is trained on a different bootstrap sample of the training data, and at each split is allowed to consider only a random handful of the predictors rather than all of them. Averaging the predictions across the forest may result in more accurate predictions than a single model. It complements the regression approaches because it can pick up nonlinear effects and interactions, for example, the idea that high scoring matters much more when paired with high minutes, without us having to specify them in advance.

The main hyperparameter, `mtry`, controls how many predictors each split is allowed to consider. We pick between $\lfloor\sqrt{p}\rfloor$ and $\lfloor p/3\rfloor$ using out-of-bag (OOB) error, which is essentially a free, built-in validation score: every prediction is averaged only over the trees that did not see that player during training. Variable importance is summarized by mean decrease in Gini impurity, which ranks predictors by how much they help the trees separate the four outcome classes. Held-out test accuracy and macro-averaged F1 are then used for direct comparison against the two regression-based models.

5 Results

5.1 Jiwon Park - Multinomial Logistic Regression

The multinomial logistic regression serves as the interpretable baseline multiclass model. Coefficient estimates indicate that stronger early scoring and more playing time are positively associated with stronger career outcomes, while worse draft position works in the opposite direction. For the sports question at hand, this means that highly drafted players who fail to earn early minutes or produce efficiently are much more likely to slide toward disappointing career outcomes than toward starter or star trajectories.

The bootstrap-based uncertainty summaries show that the model's conclusions are directional rather than precise point estimates. Early scoring, minutes, and draft position repeatedly appear as the strongest predictors separating successful early picks from flops across all 500 bootstrap resamples, lending confidence to those directions even when exact coefficient magnitudes vary.

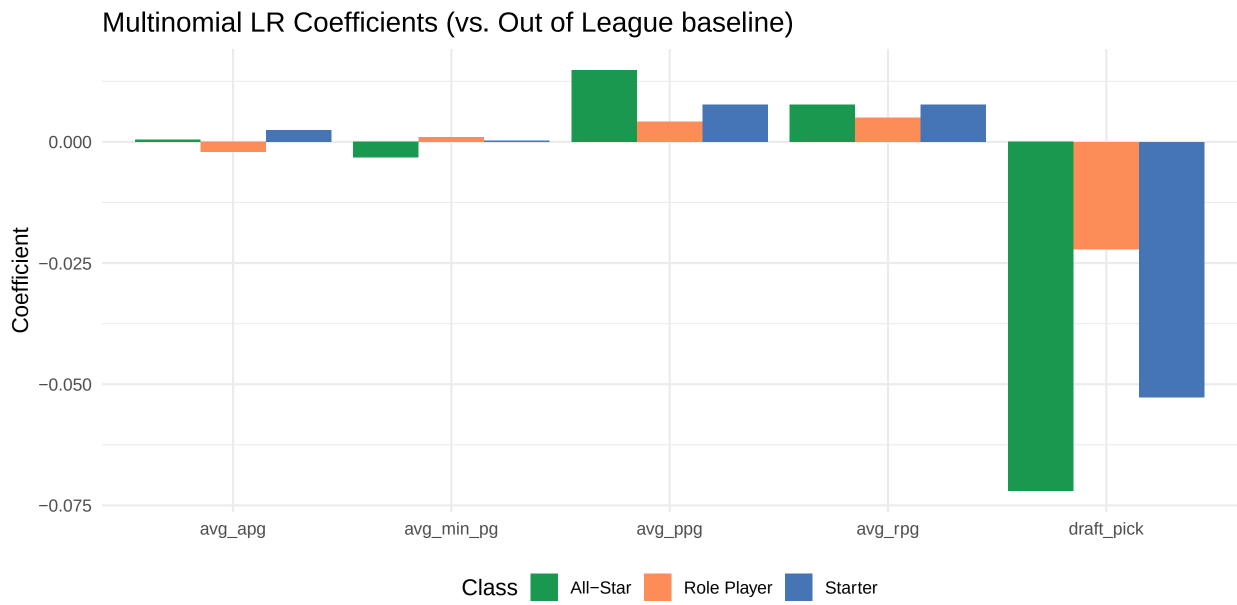

Figure 4: Multinomial logistic regression coefficients for key predictors, relative to Out of League baseline. Error bars omitted here; bootstrap CIs were computed for all coefficients.

5.2 Andy Ouyang - Bayesian Logistic Model

The Bayesian model produces similar substantive conclusions to the multinomial logistic regression, but with a more direct representation of predictive uncertainty. Rather than only assigning one class to a player, the model returns a distribution over possible career outcomes. That is valuable for this project because many highly drafted players are borderline cases early on rather than obvious stars or obvious flops. Since all R-hat values were below 1.05, there is no evidence of convergence issues within the MCMC chains.

Interestingly, the draft pick coefficient is positive for Starter and All-Star outcomes and negative for Role Player outcomes, all relative to the Out-of-League baseline. This seems rather counterintuitive, but indicates that conditional on early-career performance and inclusion in the sample, increases in draft pick number shift probability mass away from the Role Player category toward higher-tier outcomes. This is best understood as a data artifact rather than a real effect, driven by survivorship bias (only the best late picks accumulate enough seasons to appear in the sample) and the fact that the early-career performance variables already absorb most of the true draft quality signal, leaving the draft pick coefficient to pick up noise.

Scoring points per game is the clearest predictor of stardom as players who scored early either become All-Stars or wash out and don't tend to settle into role player careers. Rebounds follow an interesting non-linear pattern, where it positively predicts starter outcomes but is negatively associated with both role player and All-Star tiers. This likely reflects positional differences as the tall and strong players tend to rack up rebounds and reach starter-level roles, whereas All-Star players are more often guards and wings, for whom rebounding is less central to their game. Assists, minutes played, and field goal percentage are all predictors that indicate survival, as they are are most positive for Role Player and shrink toward zero or are negative for higher tiers. Assists is the weakest signal since the intervals all include 0. As such, assists help bench players survive but don't predict becoming a starter or star and is overall ambiguous. Minutes played and field goal percentage show overall positive credible intervals for role players, which serve as a strong signal for staying in the league as a role player, with field percentage having the largest coefficient by far.

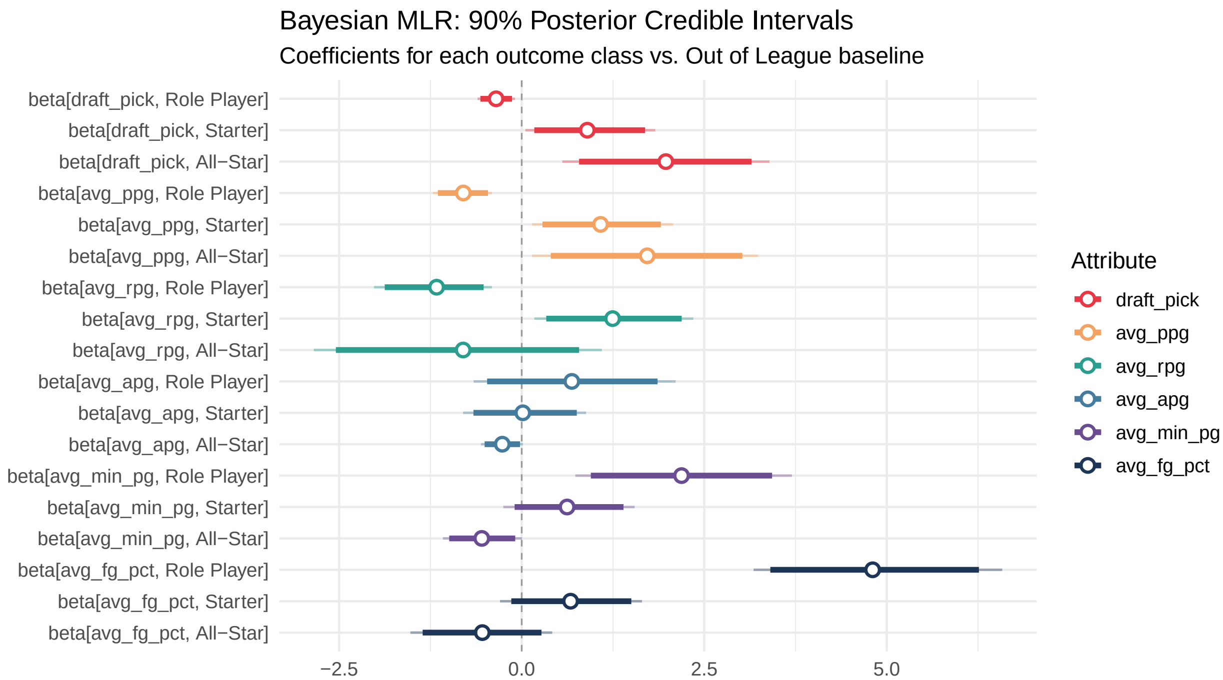

Figure 5: Bayesian MLR: 90% posterior credible intervals for coefficients in each outcome class versus the Out of League baseline.

The profile comparison below shows that both place most of their posterior mass on Out of League, reflecting the overall difficulty of sustaining an NBA career. The lottery pick profile shifts modest probability toward Role Player relative to the late second-round profile, but neither profile produces meaningful probability for Starter or All-Star outcomes, underscoring that even strong early-career performance is an uncertain predictor of sustained success. Overall, the Bayesian model shows that early-career statistics provide meaningful but uncertain signals of long-run career outcomes.

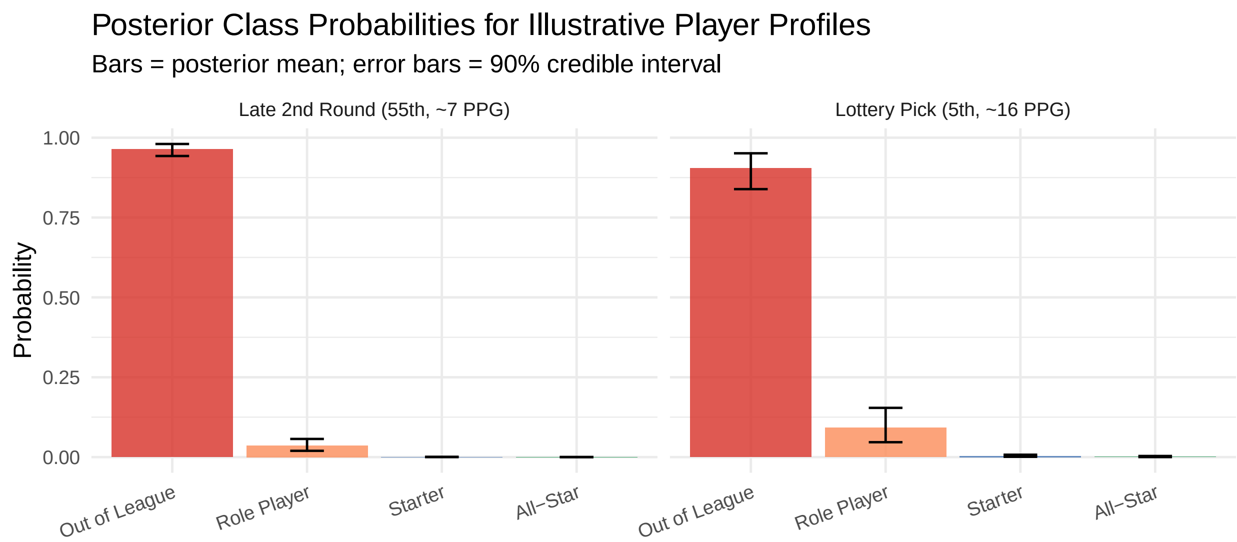

Figure 6: Posterior class probabilities for two illustrative player profiles. Bars show posterior means; error bars show 90% credible intervals.

5.3 Stanley Ou - Random Forest

The random forest gives a flexible benchmark and broadly agrees with the regression-based models about which inputs matter most. Variable importance confirms that early scoring, playing time, and draft position are again the central predictors. This consistency is meaningful and suggests that the main conclusions about career outcome prediction from early statistics isn't dependent on model choice.

Although RFs are less interpretable than regression models, we can use the variable importance plot below to understand which factors led to certain prediction outcomes. Two predictors stand out well above the rest: average points per game and average minutes per game in the first three seasons. They are essentially tied at the top, which fits the intuition that early scoring and early playing time together capture most of what teams are signaling about a young player's role and trajectory. Below them, the next tier is made up of the other counting stats such as steals, rebounds, and assists. It is interesting to see that almost all box score stats mattered for the model.

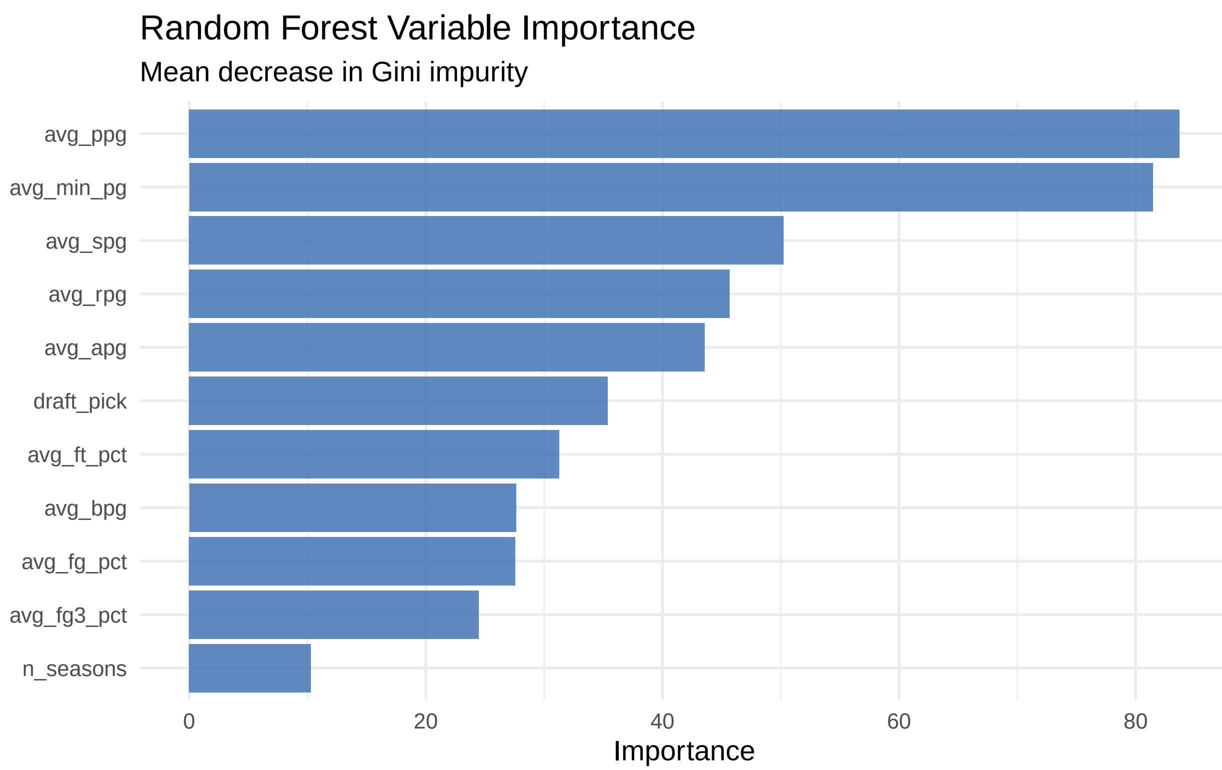

Figure 7: Random forest variable importance (mean decrease in Gini impurity).

Draft pick lands in the middle of the pack rather than at the top. This is consistent with what the regression models suggested: draft position carries information about a prospect, but once early-career production is also in the model, much of the signal in the draft pick is already being expressed through the player's actual on-court statistics. The shooting percentage variables (free throw, field goal, three-point) sit a step lower in importance than the counting stats, which makes sense for tier prediction since efficiency doesn't factor in how long a player plays. Finally, `n_seasons` is by far the least important predictor, which is reassuring since it confirms that whether we observed one, two, or three seasons of a young player is mostly absorbed by the actual statistics from those seasons rather than acting as an independent signal.

5.4 Model Comparison

Table 1: Test-set performance across all three models

Model	Accuracy	Macro_F1
MLR	0.655	0.560
RF	0.678	0.526
Bayes	0.672	0.577

At the aggregate level, the three models land in a narrow band. The random forest has the highest accuracy (0.678), the Bayesian model is essentially tied with it (0.672), and the multinomial logistic regression trails slightly (0.655). Macro-F1, which weights all four classes equally rather than by frequency, tells a different story: the Bayesian model leads (0.577) while the random forest is actually the worst on this metric (0.526). That gap between accuracy and macro-F1 reveals how the random forest is worse at classifying the rarer classes.

Table 2: Per-class precision for each model.

Class	MLR	RF	Bayes
Class: Out of League	0.768	0.759	0.754
Class: Role Player	0.645	0.667	0.660
Class: Starter	0.400	0.579	0.500
Class: All-Star	0.600	0.250	0.600

Table 3: Per-class recall for each model.

Class	MLR	RF	Bayes
Class: Out of League	0.741	0.759	0.741
Class: Role Player	0.759	0.785	0.785
Class: Starter	0.276	0.379	0.310
Class: All-Star	0.375	0.125	0.375

The per-class precision and recall tables make that tradeoff explicit. On the two majority classes (`Out of League` and `Role Player`), all three models perform similarly, with precision and recall both sitting in the 0.65–0.79 range. The interesting disagreement appears in the smaller classes. For `Starter`, the random forest has the best precision (0.579) and recall (0.379), but recall is still below 40% across all three models. This suggests that most players who actually become Starters get pushed into adjacent tiers by every model. For `All-Star`, we see the opposite where the multinomial LR and Bayesian model both reach 0.60 precision with 0.375 recall, while the random forest collapses to 0.25 precision and 0.125 recall, meaning it correctly flags only a handful of true All-Stars in the held-out set. This `All-Star` shortfall is what drags the random forest's macro-F1 down. For an applied use case, the choice between models depends on whether identifying rare stars matters more than overall classification accuracy.

5.5 Player-Level Predictions

To illustrate how the models behave on specific players rather than only in aggregate, we generate predictions from all three models on a stratified sample of held-out players and compare them to each player's actual career outcome. This serves as a sanity check and is also interesting to see if the models' predictions matches our preconceived understanding of these players.

Table 4: Predicted vs. actual career outcomes for a stratified sample of 16 held-out players (4 per outcome tier).

Player	Draft Pick	Actual	MLR Pred	Bayes Pred	RF Pred
Steven Hunter	15	Out of League	Out of League	Out of League	Out of League
John Jenkins	23	Out of League	Out of League	Out of League	Role Player
David Noel	39	Out of League	Role Player	Role Player	Out of League
Antonis Fotsis	47	Out of League	Out of League	Out of League	Out of League
Martell Webster	6	Role Player	Role Player	Role Player	Role Player
Charlie Villanueva	7	Role Player	Role Player	Role Player	Role Player
Roy Hibbert	17	Role Player	Role Player	Role Player	Role Player
Jared Sullinger	21	Role Player	Role Player	Role Player	Role Player
Emeka Okafor	2	Starter	Role Player	Role Player	Role Player
Antoine Walker	6	Starter	All-Star	All-Star	All-Star
Ron Harper	8	Starter	Role Player	Role Player	Role Player
Carlos Boozer	34	Starter	Role Player	Role Player	Role Player
Tim Duncan	1	All-Star	All-Star	All-Star	Starter
LaMarcus Aldridge	2	All-Star	Starter	Starter	Role Player
Paul Pierce	10	All-Star	All-Star	All-Star	All-Star
Paul George	10	All-Star	Starter	Starter	Starter

At a high level, we see that across all models, players are typically classified at least one tier within their true label. I.e we don't see any all stars classified as out of league. The per-player view reinforces the patterns visible in the precision and recall tables. The models broadly agree with one another more often than they disagree, and when they err they tend to err in the same direction usually toward `Out of League` or `Role Player`, the two most common classes.

6 Discussion

The main conclusion we found is that early career NBA production contains meaningful information about long-run career outcomes. Across all three models, the most consistent signals are early scoring production and early playing time. Draft position alone does not determine success. From our EDA, we found substantial overlap across outcome tiers but combining with early performance statistics separates players who wash out from those who become reliable starters or stars. The consistency of this result across a multinomial logistic regression, a Bayesian multinomial model, and a random forest strengthens confidence in the conclusion.

One key limitation is the construction of the career outcome labels. The thresholds used to define `All-Star`, `Starter`, `Role Player`, and `Out of League` are approximations based on career points per game and minutes per game rather than official accolade records. This means some borderline players may be misclassified, and the `All-Star` class in particular is small and may not perfectly capture true All-Star selections. A second limitation is survivorship bias: the data only includes players who appeared in at least ten games in a season, which removes very short-tenured players entirely and may distort the `Out of League` class. Third, the predictor set is limited to traditional box score statistics; advanced metrics such as VORP or BPM, or physical measurements from the draft combine, could improve prediction.

Future work could address these limitations by pulling official All-Star appearance records and using them to define outcome labels more precisely. Incorporating player position as a covariate would also be natural, since the tradeoffs between scoring, rebounding, and assists differ substantially across guards, forwards, and centers. Finally, extending the prediction window, for example, asking what happens after seasons 4 through 6 rather than the first three alone could help model the late bloomer versus early starter distinction, which is another important dimension of the career outcome prediction problem.